Ten Essential Concepts in Statistics

A concise guide for introductory students

Abstract

Statistics provides a language for learning from data while acknowledging uncertainty. These notes develop ten foundational ideas in a logical sequence: classifying data, describing it numerically and graphically, modeling chance, making inferences, and studying relationships. Definitions, worked examples, assumptions, and interpretations are emphasized throughout.

Contents

1 Types of Data

A *variable* is a characteristic recorded for each observational unit; its recorded values form data. The kind of variable determines which summaries, graphs, and statistical procedures are meaningful. Before classifying variables, identify where the data came from and what population the data are meant to represent.

1.1 Data structure and sources

A *population* is the complete group or process about which conclusions are desired. A *sample* is the subset actually observed. An *observational unit* is the person, object, time period, or case on which measurements are recorded. A *data value* is one recorded value of one variable for one unit. A rectangular *data matrix* usually puts observational units in rows and variables in columns. A *sampling frame* is the list or mechanism from which the sample is selected; if the frame omits relevant units, generalizability is limited.

Table 1 is a hypothetical classroom data matrix. If the research question concerns all students in a university statistics course, the population might be all enrolled students, and a possible sampling frame is the registrar's course roster.

Table 1: A hypothetical five-student data matrix.

Student	Major	Academic year	Study hours	Exam score	Attended review?
A	Economics	First	2	58	No
B	Biology	Second	4	71	Yes
C	History	First	3	65	No
D	Mathematics	Third	6	82	Yes
E	Economics	Second	5	76	Yes

In this table, major and review attendance are nominal categorical variables, academic year is ordinal, and study hours and exam score are quantitative. Study hours are continuous in an ideal measurement model but may be rounded when recorded; exam score is quantitative on the test scale.

1.2 From data to conclusions

Simple random sampling gives each possible sample of the desired size the same chance of selection. *Stratified sampling* first divides the population into important groups, such as academic years, and then samples within each group. *Cluster sampling* randomly selects natural groups, such as sections or dormitories, and observes units within selected groups. *Convenience sampling* uses units that are easy to reach; it is often fast but can be badly biased.

An *observational study* records variables without assigning treatments. A *randomized experiment* assigns treatments by a chance mechanism. Random sampling supports generalization to a suitably represented population; random assignment supports causal conclusions under appropriate experimental controls. Neither mechanism automatically guarantees the other.

Selection bias occurs when the way units enter the sample systematically favors some outcomes. *Nonresponse bias* occurs when selected units who do not respond differ from responders. *Measurement bias* occurs when the measurement process systematically overstates or understates values.

Table 2: Design features and the conclusions they support.

Design feature	Main support	Limitation
Random sample from a good frame	Generalizing sample patterns to the frame population	Does not by itself prove cause and effect
Random assignment in an experiment	Comparing treatments causally, if controls are sound	Does not by itself ensure the subjects represent a wider population
Convenience or volunteer sample	Quick description of responding units	Vulnerable to selection bias and weak generalizability
Observational comparison	Association in naturally occurring data	Confounding and reverse direction can remain

Confounding occurs when an outside variable is related to both an explanatory variable and a response, making a causal explanation ambiguous. For example, a volunteer survey about sleep in an early-morning class may overrepresent students willing to attend early classes and underrepresent absent or sleep-deprived students. A random sample from the registrar list would usually support broader generalization, although nonresponse and measurement quality still need attention.

1.3 Qualitative and quantitative variables

Qualitative (categorical) data place units into groups. Their labels need not be numbers. *Quantitative* data record numerical amounts or counts for which arithmetic has a meaningful interpretation.

Table 3: The principal data types.

Type	Definition	Examples
Qualitative	Category or quality rather than a numerical magnitude	Favorite sport; a movie rating; major code 01
Quantitative	Numerical count or measurement	Number of goals; height; temperature; exam score

1.3.1 Nominal and ordinal categories

Nominal categories have no inherent ranking. For example, favorite sport might be hockey, golf, or soccer; no category is intrinsically above another. Ordinal categories have an order, but gaps between categories are not necessarily equal. Movie ratings *bad*, *fair*, and *good* are ordered, yet the distance from bad to fair is not a measurable numerical interval. Numeric codes can still represent categorical labels: jersey numbers, identification numbers, and major codes are not quantitative measurements. Arithmetic with ordinal codes, such as averaging satisfaction ratings coded 1 through 5, requires caution unless an interval-scale interpretation is defensible.

1.3.2 Discrete and continuous quantities

A discrete variable has a finite or countably infinite set of possible values, often arising from counting. Numbers of goals and students are discrete: fractional students are not possible. A continuous

Table 4: Two forms of qualitative data.

Scale	Ordering?	Appropriate comparisons
Nominal	No natural order	Equal/not equal; category frequencies
Ordinal	Natural order	Equal/not equal and before/after; ranks

variable can, in an ideal measurement model, take values over a finite interval, a semi-infinite interval, the whole real line, or a union of intervals. Height and temperature are continuous even when an instrument rounds their recorded values, because continuous measurement models are idealizations of the underlying quantity.

Table 5: Two forms of quantitative data.

Type	Possible values	Typical origin
Discrete	Finite or countably infinite set	Counting, such as goals
Continuous	Values in intervals under a measurement model	Measuring, such as height

Key points

Data type guides numerical summaries, graphs, and procedures. Study design guides what conclusions are justified: random sampling concerns representativeness, while random assignment concerns causation.

2 Measures of Central Tendency

A *measure of central tendency* is a numerical description of a typical or central value in a distribution. The mean, median, and mode capture different notions of center.

2.1 Mean

For observations $x_1, x_2, \ldots, x_n$, the sample mean is

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i. \tag{1}$$

Here x_i is observation i, n is the sample size, $\sum$ denotes addition over all observations, and $\bar{x}$ is the sample mean. The population analogue is

$$\mu = \frac{1}{N}\sum_{i=1}^{N} x_i,$$

where N is population size and μ is the population mean.

For seven illustrative test scores

$$72,\ 78,\ 80,\ 85,\ 85,\ 86,\ 86,$$

the sum is 572, so

$$\bar{x} = \frac{572}{7} = 81.714\ldots \approx 81.7 \text{ points.}$$

The mean is a balance point and uses every value, so an extreme score can move it substantially.

2.2 Weighted and frequency means

When observations have weights w_i, the weighted mean is

$$\bar{x}_w = \frac{\sum_i w_i x_i}{\sum_i w_i}. \tag{2}$$

For a frequency table with value x_i occurring f_i times,

$$\bar{x} = \frac{\sum_i f_i x_i}{\sum_i f_i}. \tag{3}$$

For quiz scores $6, 7, 8, 9$ with frequencies $1, 2, 3, 2$, respectively,

$$\bar{x} = \frac{1(6) + 2(7) + 3(8) + 2(9)}{1 + 2 + 3 + 2} = \frac{62}{8} = 7.75.$$

Table 6: A reproducible frequency table for a mean.

Score x_i	Frequency f_i	Product $f_i x_i$	Interpretation
6	1	6	one score of 6
7	2	14	two scores of 7
8	3	24	three scores of 8
9	2	18	two scores of 9
Total	8	62	$\bar{x} = 62/8$

2.3 Median and mode

After sorting n observations, the *median* is the middle observation when n is odd and the mean of the two middle observations when n is even. For the ordered odd-size data

$$62,\ 70,\ 70,\ 85,\ 90,$$

the median is the third value, 70. For the ordered even-size data

$$62,\ 70,\ 85,\ 90,$$

the median is $(70 + 85)/2 = 77.5$.

The *mode* is the most frequent value or category. The data $62, 70, 70, 85, 90$ have mode 70. The data $1, 2, 3, 4$ have no mode if a mode is required to occur more often than the rest. The data $1, 1, 2, 3, 3, 4$ are bimodal, with modes 1 and 3.

Table 7: Choosing a measure of center.

Measure	Definition	Strength	Caution
Mean	Arithmetic average	Uses every value	Sensitive to outliers and skew
Median	Middle ordered value	Resistant to outliers	Ignores magnitudes away from center
Mode	Most frequent value	Works for categories	May be absent or nonunique

Common cautions

Always report units. A mean need not be an observed value. For a strongly skewed quantitative distribution, the median often better represents a typical observation. Right-skewed distributions often have mean greater than median, and left-skewed distributions often have mean less than median, but this is a tendency rather than a universal theorem. For nominal data, the mode is the only one of these three measures that applies. Center alone is incomplete, motivating spread.

3 Measures of Spread

A *measure of spread* describes how far observations lie from one another or from their center. Two data sets can share a mean yet differ greatly in variability.

3.1 Range, percentiles, quartiles, and IQR

For ordered data, the range and interquartile range (IQR) are

$$\text{range} = x_{\max} - x_{\min}, \tag{4}$$
$$\text{IQR} = Q_3 - Q_1, \tag{5}$$

where $x_{\max}$ and $x_{\min}$ are the maximum and minimum, Q_1 is the first quartile, Q_2 is the median, and Q_3 is the third quartile. A percentile is a location in an ordered distribution; roughly, the pth percentile has about $p\%$ of observations at or below it. The *five-number summary* is

$$\text{minimum}, \quad Q_1, \quad \text{median}, \quad Q_3, \quad \text{maximum}.$$

These notes use the median-of-halves convention: after finding the median, Q_1 is the median of the lower half and Q_3 is the median of the upper half, excluding the overall median when n is odd. Software may use other legitimate percentile conventions, so the convention should be stated.

For ordered study hours

$$1,\ 2,\ 4,\ 5,\ 5,\ 6,\ 7,\ 10,$$

the sum is 40, so $\bar{x} = 40/8 = 5$ hours. The range is $10 - 1 = 9$ hours. Using the median-of-halves convention,

$$Q_1 = \frac{2+4}{2} = 3, \qquad Q_2 = \frac{5+5}{2} = 5, \qquad Q_3 = \frac{6+7}{2} = 6.5,$$

and therefore $\text{IQR} = 6.5 - 3 = 3.5$ hours. The middle half of the observations spans 3.5 hours.

3.2 Outlier fences and box-plot conventions

A common rule flags observations below or above the fences

$$Q_1 - 1.5\,\mathrm{IQR}, \qquad Q_3 + 1.5\,\mathrm{IQR}. \tag{6}$$

For the study-hour data, the fences are

$$3 - 1.5(3.5) = -2.25, \qquad 6.5 + 1.5(3.5) = 11.75.$$

No observation is outside these fences. A min–max box plot would use whiskers at 1 and 10. A Tukey-style box plot also uses whiskers at the most extreme non-outlying observations; here those are again 1 and 10. If a value such as 18 hours were added, the quartiles and fences would need to be recomputed; outlying values would be plotted separately rather than used as Tukey whisker endpoints.

3.3 Variance, standard deviation, and standardized values

Population variance and standard deviation are

$$\sigma^2 = \frac{1}{N}\sum_{i=1}^{N}(x_i - \mu)^2, \qquad \sigma = \sqrt{\sigma^2}. \tag{7}$$

Sample variance and standard deviation are

$$s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2, \qquad s = \sqrt{s^2}. \tag{8}$$

Dividing by $n-1$ makes s^2 an unbiased estimator of population variance under random sampling. Once $\bar{x}$ is estimated, the deviations sum to zero, leaving $n-1$ independently variable deviations.

For the study-hour sample, the squared deviations from 5 are

$$16,\ 9,\ 1,\ 0,\ 0,\ 1,\ 4,\ 25,$$

whose sum is 56. Hence

$$s^2 = \frac{56}{8-1} = 8, \qquad s = \sqrt{8} = 2.828\ldots \approx 2.8 \text{ hours}.$$

Standard deviation is a root-mean-square scale of deviations from the mean, not the mean absolute distance from the mean. There is no universal rule that a fixed percentage of arbitrary data lies within one standard deviation; about 68% within one standard deviation is a normal-model result.

A descriptive sample standardized value is

$$z_i = \frac{x_i - \bar{x}}{s}. \tag{9}$$

For the 10-hour observation, $z = (10-5)/2.828 \approx 1.77$, so it is about 1.77 sample standard deviations above the sample mean.

Key points

Variance has squared units; standard deviation has the original units. Pair mean with standard deviation for roughly symmetric data and median with IQR for skewed data. Graphs in Section 4 reveal shape that no single spread statistic can show.

Table 8: Comparison of spread measures.

Measure	Formula	Uses	Sensitivity
Range	$x_{\max} - x_{\min}$	Full span	Very sensitive to extremes
IQR	$Q_3 - Q_1$	Middle 50%	Resistant to extremes
Standard deviation	Root mean squared scale of deviations	Spread about mean	Sensitive to extremes

4 Graphical Representations of Data

A *statistical graph* maps data values or categories to visual marks so that frequencies, proportions, distributional shape, and unusual values can be seen.

4.1 Categorical displays and frequency tables

For favorite subject, the frequencies are Mathematics 12, Science 9, English 3, and History 6, for a total of 30. A frequency table gives counts, while a relative-frequency table divides each count by the total.

Table 9: Favorite-subject frequency and relative-frequency table.

Subject	Frequency	Relative frequency
Mathematics	12	$12/30 = 0.40 = 40\%$
Science	9	$9/30 = 0.30 = 30\%$
English	3	$3/30 = 0.10 = 10\%$
History	6	$6/30 = 0.20 = 20\%$
Total	30	$1.00 = 100\%$

A bar chart gives each category a separated bar with height equal to its frequency or relative frequency. A pie chart uses sector proportions; for example Mathematics occupies 40% of the circle. Cumulative relative frequency is meaningful for ordered categories or quantitative intervals, but not for unordered subjects unless an order is imposed for a special purpose.

4.2 Quantitative displays and distribution shape

A histogram divides a quantitative scale into intervals (bins) and uses touching bars to display frequencies or densities. Bin widths and boundaries must be reported because they influence appearance. A dotplot places one dot for each observation along a number line and is useful for small samples; a stem-and-leaf plot is another compact hand display. A box plot summarizes the five-number summary and is especially effective for comparing groups.

Describe a distribution

For a quantitative distribution, describe shape, center, spread, and unusual features in context. Shape includes symmetry, right or left skew, unimodality or bimodality, clusters, gaps, and outliers or contextual anomalies.

Compact dotplot sketches can show the main shape ideas without extra packages:

Roughly symmetric	1: ● 2: ● ● 3: ● ● ● 4: ● ● 5:●
Right-skewed	1: ● ● ● 2: ● ● 3: ● 6:●
Bimodal	1: ● ● 2: ● 5: ● 6: ● ●

The first sketch is balanced around its center, the second has a long right tail, and the third has two clusters separated by a gap.

For student heights, the supplied five-number summary is

$$151, \quad 158.5, \quad 163.5, \quad 169, \quad 178.$$

Thus the box extends from $Q_1 = 158.5$ to $Q_3 = 169$, a line marks the median 163.5, and whiskers reach the minimum 151 and maximum 178 under the simple min–max convention. Under a Tukey convention, outlier fences require raw data or enough information to identify non-outlying extremes.

Table 10: Common statistical displays.

Display	Data type	Main purpose	Important caution
Bar chart	Categorical	Compare counts/proportions	Bars are separated; use a common baseline
Pie chart	Categorical	Show parts of one whole	Poor for many or similar-sized categories
Dotplot	Quantitative	Show individual values in small samples	Can become crowded for large samples
Histogram	Quantitative	Show shape, center, and spread	Conclusions depend on bins
Box plot	Quantitative	Show quartiles and compare groups	Conceals modes and detailed shape
Scatterplot	Paired quantitative	Show form, direction, strength, clusters, and outliers	Association need not be linear or causal

Common cautions

Do not use a bar chart as though categories were a continuous numerical scale, or a histogram for category labels. Axes, units, binning, and the sample represented must be clear. Visualized relative frequencies lead naturally to probability.

5 Probability Fundamentals

Probability is a number from 0 to 1 that quantifies uncertainty about an event: 0 means impossible and 1 means certain. Equivalent forms include

$$0.7 = 70\% = \frac{7}{10}.$$

5.1 Experiments, outcomes, events, and axioms

A random experiment has an uncertain result. Its *sample space* S is the set of possible outcomes, and an *event* A is a subset of S. The basic probability axioms are

$$0 \le P(A) \le 1, \qquad P(S) = 1, \tag{10}$$

and if A and B are disjoint events, then

$$P(A \cup B) = P(A) + P(B).$$

When all elementary outcomes are equally likely,

$$P(A) = \frac{\text{number of outcomes favorable to } A}{\text{total number of outcomes in } S}. \tag{11}$$

This favorable-over-total formula is not valid for unequal elementary outcomes unless the outcomes have first been weighted appropriately. For a fair six-sided die, $P(\text{three}) = 1/6$.

The complement and general addition rules are

$$P(A^c) = 1 - P(A), \tag{12}$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B), \tag{13}$$

where A^c means "not A," $A \cup B$ means "A or B (or both)," and $A \cap B$ means "both."

5.2 Conditional probability, multiplication, and independence

Conditional probability is

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}, \qquad P(B) > 0. \tag{14}$$

Rearranging gives the general multiplication rule

$$P(A \cap B) = P(A \mid B)P(B) = P(B \mid A)P(A). \tag{15}$$

Events A and B are independent if knowing one occurred does not change the probability of the other; equivalently,

$$P(A \cap B) = P(A)P(B). \tag{16}$$

Independent fair die rolls therefore give

$$P(\text{two consecutive threes}) = \frac{1}{6} \cdot \frac{1}{6} = \frac{1}{36}.$$

Mutually exclusive events cannot occur together; independent events do not influence one another. They are not synonymous. On one die roll, $A =$ "roll a 1" and $B =$ "roll a 2" are mutually exclusive, but because $P(A \cap B) = 0 \neq (1/6)(1/6)$, they are not independent. On two die rolls, $A =$ "first roll is 1" and $B =$ "second roll is 1" are independent but not mutually exclusive, because both can occur.

5.3 Total probability and Bayes' theorem

If B and B^c split the sample space, then

$$P(A) = P(A \mid B)P(B) + P(A \mid B^c)P(B^c). \tag{17}$$

Bayes' theorem reverses a conditional probability:

$$P(B \mid A) = \frac{P(A \mid B)P(B)}{P(A)}. \tag{18}$$

Suppose a rare condition affects 1% of a population. A screening test is positive for 90% of people with the condition and for 5% of people without it. For a hypothetical group of 10,000 people, the expected counts are:

Table 11: A 2×2 table illustrating Bayes' theorem and base rates.

Group	Positive test	Negative test	Total
Condition	90	10	100
No condition	495	9405	9900
Total	585	9415	10000

Thus $P(\text{positive}) = 585/10000 = 0.0585$. The reversed conditional probability is

$$P(\text{condition} \mid \text{positive}) = \frac{90}{585} = 0.1538\ldots \approx 15.4\%.$$

Even with a sensitive test, most positive tests are false positives because the condition is rare; this is the base-rate effect.

5.4 Theoretical and experimental probability

Experimental probability, or relative frequency, estimates probability from observed repetitions:

$$\widehat{P}(A) = \frac{\text{number of observed occurrences of } A}{\text{number of trials}}. \tag{19}$$

If two threes occur in 20 rolls, $\widehat{P}(\text{three}) = 2/20 = 0.10$. This differs from $1/6 \approx 0.1667$ because short-run samples fluctuate; with many independent trials under stable conditions, relative frequency tends to approach the true probability.

5.5 Counting methods used in probability

Counting methods

For a positive integer n, $n! = n(n-1)\cdots 2 \cdot 1$ and $0! = 1$. Permutations count ordered arrangements: $P(n, r) = n!/(n-r)!$. Combinations count unordered selections: $\binom{n}{r} = n!/[r!(n-r)!]$. The binomial formula uses combinations because it counts which k of the n trials are successes, not the order in which the selected positions are named.

Key points

Probability rules become distribution rules when outcomes are assigned numerical values. Discrete distributions use probability masses; continuous distributions use density and area.

6 Discrete Probability Distributions

A *discrete random variable* X maps outcomes to a finite or countably infinite numerical set. Its probability mass function (PMF) is $p(x) = P(X = x)$, with $p(x) \geq 0$ and $\sum_x p(x) = 1$. Its cumulative distribution function (CDF) is

$$F(x) = P(X \leq x) = \sum_{t \leq x} p(t). \tag{20}$$

6.1 Expected value, variance, and rules

For possible values x,

$$\mu_X = \mathrm{E}(X) = \sum_x x p(x), \tag{21}$$

$$\mathrm{Var}(X) = \sum_x (x - \mu_X)^2 p(x), \qquad \sigma_X = \sqrt{\mathrm{Var}(X)}. \tag{22}$$

Useful rules include

$$\mathrm{E}(a + bX) = a + b\,\mathrm{E}(X), \tag{23}$$

$$\mathrm{Var}(a + bX) = b^2\,\mathrm{Var}(X), \tag{24}$$

and, when X and Y are independent,

$$\mathrm{Var}(X + Y) = \mathrm{Var}(X) + \mathrm{Var}(Y). \tag{25}$$

6.2 Bernoulli and binomial distributions

A Bernoulli random variable records one trial with success probability p: $X = 1$ for success and $X = 0$ for failure. A binomial experiment repeats this structure a fixed number n of independent trials, each with the same success probability p. If X counts successes, then

$$X \sim \mathrm{Binomial}(n, p), \qquad P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}, \qquad k = 0, \ldots, n. \tag{26}$$

Also $\mathrm{E}(X) = np$ and $\mathrm{Var}(X) = np(1 - p)$.

Let success mean rolling a six, in three independent fair die rolls. Then

$$X \sim \mathrm{Binomial}\left(3, \frac{1}{6}\right).$$

Applying (26) produces the complete distribution and CDF:

The numerators in the PMF sum to 216, verifying total probability 1. The expected number of sixes is $np = 3(1/6) = 0.5$; this long-run average need not be an attainable count in one experiment. A tail probability can be computed from the complement:

$$P(X \geq 1) = 1 - P(X = 0) = 1 - \frac{125}{216} = \frac{91}{216} \approx 0.4213.$$

Table 12: PMF and CDF for $X \sim \text{Binomial}(3, 1/6)$.

k	$P(X = k)$	$F(k) = P(X \leq k)$
0	125/216	125/216
1	75/216	200/216
2	15/216	215/216
3	1/216	216/216 = 1

6.3 The Poisson model for counts

A Poisson random variable models counts in a specified interval of time, space, area, or volume under a constant-rate model. If the expected count in the interval is $\lambda > 0$, then

$$P(X = k) = e^{-\lambda} \frac{\lambda^k}{k!}, \qquad k = 0, 1, 2, \ldots, \tag{27}$$

where $\lambda = \mathrm{E}(X) = \mathrm{Var}(X)$. Under the model, counts in disjoint intervals are independent and the expected count is proportional to interval length. Dependence, clustering, bounded counts, or changing rates can invalidate the model.

If calls arrive at an average rate of 3 per hour, then for one hour $X \sim \text{Poisson}(3)$ and

$$P(X = 0) = e^{-3} \approx 0.0498, \qquad P(X \geq 1) = 1 - e^{-3} \approx 0.9502.$$

For a half-hour interval under the same constant rate, the expected count is $\lambda = 1.5$ rather than 3.

Key points

A valid PMF assigns probabilities to individual values, and all masses sum to 1. Check binomial and Poisson conditions before using their formulas; sampling without replacement can violate independence, and changing rates can invalidate a Poisson model.

7 Continuous Probability Distributions

A *continuous random variable* is modeled with probabilities over intervals. Its probability density function (PDF) f satisfies $f(x) \geq 0$ and

$$P(a \leq X \leq b) = \int_a^b f(x)\, dx, \qquad \int_{-\infty}^{\infty} f(x)\, dx = 1. \tag{28}$$

Height is not itself probability: probability is area. Consequently $P(X = x) = 0$ for every individual value x. The continuous CDF is

$$F(x) = P(X \leq x) = \int_{-\infty}^{x} f(t)\, dt, \tag{29}$$

which converts density into cumulative probability.

When the integrals exist,

$$\mathrm{E}(X) = \int_{-\infty}^{\infty} x f(x)\, dx, \tag{30}$$

$$\mathrm{Var}(X) = \int_{-\infty}^{\infty} (x - \mu)^2 f(x)\, dx. \tag{31}$$

Table 13: Probability mass versus density.

Feature	PMF $p(x)$	PDF $f(x)$
Variable	Discrete	Continuous
Point value	$p(x) = P(X = x)$ may be positive	$P(X = x) = 0$
Interval probability	Sum of masses	Area $\int_a^b f(x)\,dx$
Normalization	$\sum_x p(x) = 1$	$\int_{-\infty}^{\infty} f(x)\,dx = 1$

7.1 Uniform and normal distributions

Not every continuous distribution is normal. For example, if a bus arrives uniformly during a 10-minute window, then $X \sim \text{Uniform}(0, 10)$ has density $f(x) = 1/10$ for $0 \le x \le 10$ and 0 otherwise. Therefore

$$P(2 \le X \le 5) = \int_2^5 \frac{1}{10}\,dx = 0.3.$$

A normal random variable with mean μ and standard deviation $\sigma > 0$ has density

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left[-\frac{(x-\mu)^2}{2\sigma^2}\right]. \tag{32}$$

The bell-shaped curve is symmetric about μ. Approximately 68%, 95%, and 99.7% of its area lies within one, two, and three standard deviations of μ, respectively. These empirical-rule percentages are normal-model approximations; they are not, by themselves, a test of normality.

7.2 Standardization and inverse normal calculations

The z-score of a value x in a normal population is

$$z = \frac{x-\mu}{\sigma}, \tag{33}$$

the signed number of population standard deviations between x and μ. Suppose adult height X is normally distributed with

$$\mu = 175 \text{ cm}, \qquad \sigma = 6 \text{ cm}, \qquad x = 184 \text{ cm}.$$

Then

$$z = \frac{184 - 175}{6} = 1.5.$$

If Z is standard normal and $\Phi(z) = P(Z \le z)$, a standard normal table or software gives

$$P(X \le 184) = P(Z \le 1.5) = \Phi(1.5) \approx 0.9332.$$

Thus about 93.32% of this modeled population is at most 184 cm tall.

The inverse problem starts with a probability. The 90th percentile has $z_{0.90} \approx 1.282$, so

$$x = \mu + z_{0.90}\sigma = 175 + 1.282(6) = 182.692 \text{ cm} \approx 182.7 \text{ cm}.$$

Similarly, $z_{0.975} = 1.96$ leaves 2.5% in the upper tail and 2.5% in the lower tail, giving the central 95% area used in many confidence intervals.

Common cautions

A z-score is a standardized location, not a probability. Normality should not be assumed solely because a variable is continuous. Areas under sampling distributions support the confidence intervals of the next section.

8 Confidence Intervals

A *confidence interval* is a range computed from sample data by a method designed to capture an unknown population parameter at a stated long-run rate. It combines a point estimate with a margin of error.

8.1 Sampling distributions: the bridge to inference

A *parameter* is a fixed population quantity, such as μ or p. A *statistic* is a sample quantity that varies from sample to sample. An *estimator* is the rule or random quantity used to estimate a parameter, such as $\bar{X}$ for μ; an *estimate* is the realized value, such as $\bar{x} = 9.2$.

A *sampling distribution* is the probability distribution of a statistic computed over repeated samples of the same size from the same population or process. If observations are independent with population mean μ and standard deviation σ, then for repeated samples of size n,

$$\mathrm{E}(\bar{X}) = \mu, \qquad \mathrm{SE}(\bar{X}) = \frac{\sigma}{\sqrt{n}}. \tag{34}$$

The law of large numbers says sample averages stabilize near the population mean as sample size increases. The central limit theorem says that, under appropriate conditions, the standardized sampling distribution of the sample mean becomes approximately normal as n grows; it does not say the population itself becomes normal.

Table 14: Standard deviation versus standard error.

Quantity	Describes	Changes with larger n?
Standard deviation	Variability among individual observations	Not necessarily
Standard error	Sample-to-sample variability of a statistic	Usually decreases, often like $1/\sqrt{n}$

A compact schematic is

$$\text{population distribution} \longrightarrow \text{one random sample} \longrightarrow \bar{x} \longrightarrow \text{many repeated } \bar{X}\text{'s} \longrightarrow \text{sampling distribution.}$$

This bridge is why study design from Section 1 matters for inference: biased or dependent samples can make the sampling distribution used by a procedure inappropriate.

8.2 Statistics and parameters

A statistic estimates its corresponding parameter. Its sampling variability is summarized by a standard error.

Table 15: Population quantities and sample summaries.

Concept	Population parameter	Sample statistic
Mean	μ, fixed but usually unknown	$\bar{x}$, varies between samples
Standard deviation	σ	s
Proportion	p	$\widehat{p}$

8.3 A general confidence-interval workflow

A useful workflow is: state the research question, identify the population parameter, examine sampling design and assumptions, select the procedure, calculate the estimate and uncertainty, interpret in context, and state design or model limitations. Confidence intervals estimate population parameters or effects while communicating uncertainty and precision; some, but not all, target parameters are effect-size measures.

Table 16: Common standard normal critical values.

Confidence level	z^*
90%	1.645
95%	1.960
99%	2.576

A central confidence area leaves two equal tails; for 95% confidence, 2.5% is left in each tail, so $z^* = z_{0.975} = 1.96$.

8.4 A one-sample z-interval for a mean

When population standard deviation σ is known, the interval is

$$\bar{x} \pm z^* \frac{\sigma}{\sqrt{n}} = \bar{x} \pm \mathrm{ME}, \qquad \mathrm{SE}(\bar{x}) = \frac{\sigma}{\sqrt{n}}. \tag{35}$$

This is a special case because σ is rarely known in practice.

For battery life,

$$n = 81, \qquad \bar{x} = 9.2 \text{ hours}, \qquad \sigma = 0.9 \text{ hours}, \qquad z^* = 1.96.$$

Step by step,

$$\mathrm{SE}(\bar{x}) = \frac{0.9}{\sqrt{81}} = 0.1, \qquad \mathrm{ME} = 1.96(0.1) = 0.196,$$

so

$$9.2 \pm 0.196 = (9.004, 9.396) \text{ hours}.$$

We are 95% confident that the population mean battery life lies between 9.004 and 9.396 hours. More precisely, if this sampling and interval procedure were repeated many times, approximately 95% of the constructed intervals would contain the true μ.

8.5 A one-sample t-interval for a mean

When σ is unknown and is estimated by s, use

$$\bar{x} \pm t^*_{n-1}\frac{s}{\sqrt{n}}, \qquad df = n-1. \tag{36}$$

For an illustrative sample of $n = 16$ batteries with $\bar{x} = 9.2$ hours and $s = 0.9$ hours, a 95% interval uses $df = 15$ and $t^*_{15} \approx 2.131$. Thus

$$\mathrm{SE}(\bar{x}) = \frac{0.9}{\sqrt{16}} = 0.225, \qquad \mathrm{ME} = 2.131(0.225) = 0.4795 \approx 0.480,$$

so the interval is

$$9.2 \pm 0.480 = (8.720, 9.680) \text{ hours}.$$

Under the design and model conditions, this method captures the true mean in about 95% of repeated samples.

8.6 A one-proportion interval

For a population proportion, an introductory large-sample interval is

$$\widehat{p} \pm z^*\sqrt{\frac{\widehat{p}(1-\widehat{p})}{n}}. \tag{37}$$

If 56 of 100 independent sampled students attended a review session, then $\widehat{p} = 0.56$. The success–failure counts are 56 and 44, both at least 10. A 95% interval is

$$0.56 \pm 1.96\sqrt{\frac{0.56(0.44)}{100}} = 0.56 \pm 1.96(0.04964) = 0.56 \pm 0.0973,$$

so the interval is approximately $(0.463, 0.657)$. We are 95% confident that the population proportion of students who would attend review lies between 46.3% and 65.7%, provided the sampling design supports that population conclusion.

8.7 Conditions, precision, and sample size

For means, check that the data come from a random sample or randomized process, observations are independent, and the sampling distribution of the mean is approximately normal. Approximate normality follows from a normal population or from a sufficiently large sample under the central limit theorem when extreme skew and outliers are not dominant. When sampling without replacement, a common independence check is $n \le 10\%$ of the population. For proportions, check randomization, independence, a binary outcome, and sufficient success–failure counts.

Higher confidence increases the critical value and interval width; larger n reduces standard error at rate $1/\sqrt{n}$. To plan a margin of error M for a mean using a planning value of σ,

$$n = \left(\frac{z^*\sigma}{M}\right)^2. \tag{38}$$

For a proportion,

$$n = \widehat{p}(1-\widehat{p})\left(\frac{z^*}{M}\right)^2. \tag{39}$$

If no prior estimate is available, $\widehat{p} = 0.5$ is conservative. Planned sample sizes are rounded up.

Key points

Confidence concerns the reliability of a method under repeated sampling, not the percentage of observations in the interval. A narrow interval can still be misleading when sampling is biased or assumptions fail. Confidence intervals and hypothesis tests are complementary tools for inference.

9 Hypothesis Testing

A *hypothesis test* uses sample evidence to evaluate a claim about a population. The null hypothesis H_0 states the model being tested; the alternative H_a describes the departure of interest. The same inferential workflow applies: state the question, identify the parameter, check design and conditions, compute a statistic, and interpret in context.

9.1 Logic and terminology

Choose a significance level α, the probability of rejecting H_0 when it is true (a Type I error). A Type II error is failing to reject a false H_0; power is

$$\text{power} = 1 - \beta. \tag{40}$$

A test statistic measures discrepancy between data and H_0. The p-value is computed from the null distribution of the selected test statistic in the direction or directions specified by H_a; it is the probability, assuming H_0, of a result at least as inconsistent with H_0 as the observed result. Reject H_0 when $p \le \alpha$; equivalently, reject when the test statistic falls in the rejection region determined by α. Otherwise *fail to reject* H_0. A p-value is neither the probability that H_0 is true nor an effect size.

Alternatives determine tail areas:

$$H_a : \mu > \mu_0 \quad \text{right-tailed,} \qquad H_a : \mu < \mu_0 \quad \text{left-tailed,} \qquad H_a : \mu \neq \mu_0 \quad \text{two-tailed.}$$

Power generally increases with larger true effects, larger samples, lower variability, or larger α, but increasing α also increases Type I error risk. Statistical significance and practical importance differ: a tiny effect can be significant in a very large sample, while an important effect can fail to reach significance in a small or noisy sample.

9.2 One-sample t-test for a mean

When σ is unknown, a one-sample mean test uses

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}, \qquad df = n - 1. \tag{41}$$

Using the illustrative battery sample from Section 8, test at $\alpha = 0.05$ whether the mean battery life differs from 9 hours:

$$H_0 : \mu = 9, \qquad H_a : \mu \neq 9.$$

With $n = 16$, $\bar{x} = 9.2$, and $s = 0.9$,

$$t = \frac{9.2 - 9}{0.9/\sqrt{16}} = \frac{0.2}{0.225} = 0.889, \qquad df = 15.$$

A two-tailed t calculation gives $p \approx 0.39$. Since $p > 0.05$, fail to reject H_0. The sample does not provide strong evidence that the population mean battery life differs from 9 hours; this does not prove the mean is exactly 9 hours. Conditions are the same design, independence, and shape/normality checks used for the one-sample t-interval.

9.3 One-proportion z-test

For testing $H_0 : p = p_0$, use the null-based standard error

$$z = \frac{\widehat{p} - p_0}{\sqrt{p_0(1 - p_0)/n}}. \tag{42}$$

This differs from the confidence-interval standard error, which uses $\widehat{p}$ because the interval estimates the unknown p.

For the review-attendance example, test

$$H_0 : p = 0.50, \qquad H_a : p \neq 0.50,$$

with $\widehat{p} = 56/100 = 0.56$. The null success–failure counts are $np_0 = 50$ and $n(1 - p_0) = 50$, both sufficient for the normal approximation. Then

$$z = \frac{0.56 - 0.50}{\sqrt{0.50(0.50)/100}} = \frac{0.06}{0.05} = 1.20.$$

The two-tailed p-value is approximately 0.23. At $\alpha = 0.05$, fail to reject H_0; the data do not provide strong evidence that the population attendance proportion differs from 0.50.

9.4 Chi-square goodness-of-fit test

For categorical counts, the chi-square statistic is

$$\chi^2 = \sum_{i=1}^{k} \frac{(O_i - E_i)^2}{E_i}, \tag{43}$$

where k is the number of categories, O_i is observed frequency, E_i is the expected frequency under H_0, and χ^2 is the total standardized squared discrepancy. If no parameters are estimated from the data, the degrees of freedom are $df = k - 1$.

For a hypothetical set of 60 die rolls, test

$$H_0 : \; p_1 = \cdots = p_6 = \frac{1}{6} \quad \text{against} \quad H_a : \; \text{at least one face probability differs.}$$

Each expected count is $E_i = 60(1/6) = 10$.

With $k = 6$, $df = 5$. At $\alpha = 0.05$, the critical value is approximately 11.07, and the right-tail p-value for $\chi^2 = 3.4$ is approximately 0.64. Since $3.4 < 11.07$ and $0.64 > 0.05$, fail to reject H_0. There is insufficient evidence that the die is unfair; this does not prove fairness. Pearson residuals help identify which categories contribute most to the global statistic.

Table 17: Reproducible chi-square goodness-of-fit calculation.

Face	Observed O_i	Expected E_i	Contribution $(O_i - E_i)^2/E_i$	Pearson residual
1	14	10	$16/10 = 1.6$	$4/\sqrt{10} = 1.26$
2	12	10	$4/10 = 0.4$	$2/\sqrt{10} = 0.63$
3	7	10	$9/10 = 0.9$	$-3/\sqrt{10} = -0.95$
4	8	10	$4/10 = 0.4$	$-2/\sqrt{10} = -0.63$
5	9	10	$1/10 = 0.1$	$-1/\sqrt{10} = -0.32$
6	10	10	$0/10 = 0.0$	0
Total	60	60	3.4	

9.5 Conditions and practical meaning

For one-sample mean tests, check randomization, independence, and a normal population or a sufficiently large sample without dominant skew or outliers. For one-proportion tests, check randomization, independence, binary outcomes, and null-based success–failure counts. For chi-square goodness-of-fit tests, check frequency counts from random, independent observations in mutually exclusive categories and expected counts sufficiently large; a common introductory guideline is that all $E_i \geq 5$. If probabilities or parameters are estimated from the sample, chi-square degrees of freedom must be adjusted.

Common cautions

State hypotheses before inspecting results, verify conditions, and interpret the conclusion in context. Never "accept" H_0: a nonsignificant result may reflect limited information. The next section shifts from categorical fit to quantitative relationships.

10 Regression and Correlation

Correlation quantifies the direction and strength of a linear association between two quantitative variables. *Regression* models the conditional mean or predicted response together with unexplained variation; it is not a deterministic function that exactly determines every outcome.

10.1 Correlation and scatterplots

For paired observations (x_i, y_i), Pearson's sample correlation is

$$r = \frac{1}{n-1} \sum_{i=1}^{n} \left(\frac{x_i - \bar{x}}{s_x} \right) \left(\frac{y_i - \bar{y}}{s_y} \right). \tag{44}$$

Here n is the number of pairs; $\bar{x}, \bar{y}$ and s_x, s_y are the sample means and standard deviations; and $r \in [-1, 1]$. Its sign gives the direction, while $|r|$ measures the strength of a *linear* association. It is unitless, unchanged by shifting or positive rescaling, and sensitive to outliers. Correlation does not by itself establish causation: lurking variables, reverse direction, and study design matter.

Use a scatterplot before computing or trusting r. The following hypothetical paired data will be used throughout this section.

Table 18: Hypothetical study-hours and exam-score data.

Study hours x	1	2	3	4	5	6	7	8
Score y	55	58	65	67	72	74	81	83

A compact scatterplot listing is

$$(1,55),\ (2,58),\ (3,65),\ (4,67),\ (5,72),\ (6,74),\ (7,81),\ (8,83).$$

It shows a positive, roughly linear association with no obvious isolated outlier in this small illustrative data set. Direct calculation gives

$$\bar{x} = 4.5, \quad \bar{y} = 69.375, \quad s_x \approx 2.449, \quad s_y \approx 10.070, \quad r \approx 0.9932.$$

10.2 Least-squares regression and residuals

The simple linear regression model is

$$\widehat{y} = b_0 + b_1 x, \tag{45}$$

where x is the explanatory value, $\widehat{y}$ is the predicted response, b_0 is the predicted response at $x = 0$ when that interpretation is meaningful, and b_1 is the predicted change in y for a one-unit increase in x. In simple linear regression with an intercept,

$$b_1 = r\frac{s_y}{s_x}, \qquad b_0 = \bar{y} - b_1\bar{x}. \tag{46}$$

For the paired data,

$$b_1 \approx 0.9932\left(\frac{10.070}{2.449}\right) = 4.083, \qquad b_0 = 69.375 - 4.083(4.5) = 51.000.$$

Thus

$$\widehat{y} = 51.000 + 4.083x.$$

At $x = 4$ study hours, the predicted score is

$$\widehat{y} = 51.000 + 4.083(4) = 67.333.$$

A residual is

$$e_i = y_i - \widehat{y}_i. \tag{47}$$

For the $x = 4$ observation, $e = 67 - 67.333 = -0.333$ points. Least squares chooses b_0, b_1 to minimize

$$\sum_{i=1}^{n} e_i^2. \tag{48}$$

Table 19: Fitted values and residuals for the regression example.

Observation	x_i	y_i	$\widehat{y}_i$	$e_i = y_i - \widehat{y}_i$
1	1	55	55.083	-0.083
2	2	58	59.167	-1.167
3	3	65	63.250	1.750
4	4	67	67.333	-0.333
5	5	72	71.417	0.583
6	6	74	75.500	-1.500
7	7	81	79.583	1.417
8	8	83	83.667	-0.667

10.3 Residual diagnostics, influence, and R^2

Residual plots help assess whether a line is appropriate. Random scatter around zero supports the linear form. A curved residual pattern suggests nonlinearity; changing vertical spread suggests nonconstant variance; unusually large residuals suggest response outliers; a time-ordered pattern may suggest dependence.

A response outlier is unusual in the y direction. A high-leverage observation has an unusual x value. An influential observation is one whose removal materially changes the fitted line; high leverage can make a point influential, especially when its residual is also large.

In simple linear regression with an intercept,

$$R^2 = r^2. \tag{49}$$

Here $R^2 \approx (0.9932)^2 \approx 0.9865$: about 98.65% of the observed variation in exam score is accounted for by its fitted linear relationship with study hours in this illustrative data set. This does not mean 98.65% of scores are exactly predicted or that study hours cause that share of variation.

Table 20: Association, prediction, explanation, and causation.

Use	Statistical question	Caution
Descriptive association	Do two variables move together in these data?	Does not require a causal story
Prediction	What response is expected for a given x?	Extrapolation outside the data range is risky
Explanatory modeling	How is the response related to explanatory variables?	Confounding may remain
Causal inference	What would happen under intervention?	Requires appropriate design, often random assignment

Regression checklist

Inspect the scatterplot for form, direction, strength, clusters, and unusual points. Check residuals for random scatter around zero, roughly constant variance, outliers, and dependence. Avoid extrapolation. For regression inference, also check the relevant random sampling or experimental design and any needed normal-error condition. Causal conclusions require an appropriate design; correlation or a fitted line alone is not enough.

Final Summary

Statistical work begins by identifying observational units, variables, data types, and study design. Sampling design determines what population conclusions may generalize to, while random assignment, not correlation alone, is central to causal conclusions. Center, spread, and graphs describe a sample without concealing shape, outliers, or context. Probability turns uncertain outcomes into models: discrete variables use probability masses and continuous variables use areas under density curves. Sampling distributions and standard errors connect probability models to inference. Confidence intervals estimate population parameters or effects with uncertainty, and hypothesis tests quantify evidence against a specified null model; statistical significance and practical importance are different questions. Regression and correlation describe linear association and prediction while requiring careful attention to scatterplots, residuals, influential observations, design, and causal limitations. Assumptions and diagnostic checks are part of statistical procedures, not optional afterthoughts.

Compact Formula Reference

Concept	Formula	Conditions/notes
Population/sample mean	$\mu = \dfrac{1}{N}\sum_{i=1}^{N} x_i$; $\bar{x} = \dfrac{1}{n}\sum_{i=1}^{n} x_i$	Quantitative data
Weighted/frequency mean	$\bar{x}_w = \dfrac{\sum_i w_i x_i}{\sum_i w_i}$; $\bar{x} = \dfrac{\sum_i f_i x_i}{\sum_i f_i}$	Weights/frequencies nonnegative, not all zero
Range, IQR, fences	$x_{\max} - x_{\min}$; $\mathrm{IQR} = Q_3 - Q_1$; $Q_1 \pm 1.5\mathrm{IQR}$	State quartile and box-plot convention
Variance	$\sigma^2 = \dfrac{1}{N}\sum (x_i - \mu)^2$; $s^2 = \dfrac{1}{n-1}\sum (x_i - \bar{x})^2$	s^2 estimates population variance under random sampling
Sample standardized value	$z_i = \dfrac{x_i - \bar{x}}{s}$	Descriptive; requires $s > 0$
Empirical probability	$\widehat{P}(A) = \dfrac{\#A}{\#\text{ trials}}$	Stable repeated trials for probability interpretation
Probability axioms	$0 \le P(A) \le 1$; $P(S) = 1$; disjoint additivity	Foundation for probability rules
Conditional/multiplication	$P(A \mid B) = \dfrac{P(A \cap B)}{P(B)}$; $P(A \cap B) = P(A \mid B)P(B)$	$P(B) > 0$
Independence	$P(A \cap B) = P(A)P(B)$	Special case of multiplication
Total probability/Bayes	$P(A) = P(A \mid B)P(B) + P(A \mid B^c)P(B^c)$; $P(B \mid A) = \dfrac{P(A \mid B)P(B)}{P(A)}$	B, B^c partition; $P(A) > 0$
Binomial PMF	$P(X = k) = \dbinom{n}{k} p^k (1-p)^{n-k}$	Fixed n, independent Bernoulli trials, constant p
Poisson PMF	$P(X = k) = e^{-\lambda}\lambda^k / k!$	Counts in specified interval; constant rate and independent disjoint intervals
Discrete CDF/mean/variance	$F(x) = \sum_{t \le x} p(t)$; $\mathrm{E}(X) = \sum_x x p(x)$; $\mathrm{Var}(X) = \sum_x (x - \mu_X)^2 p(x)$	Valid PMF sums to 1
Transformation rules	$\mathrm{E}(a + bX) = a + b\,\mathrm{E}(X)$; $\mathrm{Var}(a + bX) = b^2\,\mathrm{Var}(X)$; $\mathrm{Var}(X + Y) = \mathrm{Var}(X) + \mathrm{Var}(Y)$	Last rule requires independence
PDF/CDF	$P(a \le X \le b) = \int_a^b f(x)\,dx$; $F(x) = \int_{-\infty}^{x} f(t)\,dt$	$f \ge 0$ and total area 1
Continuous mean/variance	$\mathrm{E}(X) = \int_{-\infty}^{\infty} x f(x)\,dx$; $\mathrm{Var}(X) = \int_{-\infty}^{\infty} (x - \mu)^2 f(x)\,dx$	When integrals exist
Standard normal score	$z = \dfrac{x - \mu}{\sigma}$	Population/model standardization; $\sigma > 0$
Standard errors	$\mathrm{SE}(\bar{x}) = \sigma/\sqrt{n}$ or $s/\sqrt{n}$; $\mathrm{SE}(\widehat{p}) = \sqrt{\widehat{p}(1-\widehat{p})/n}$	SD describes observations; SE describes statistics
Margin of error	$\mathrm{ME} = \text{critical value} \times \mathrm{SE}$	Critical value depends on confidence level/procedure
Mean intervals	$\bar{x} \pm z^* \dfrac{\sigma}{\sqrt{n}}$; $\bar{x} \pm t^*_{n-1} \dfrac{s}{\sqrt{n}}$	z: known σ; t: unknown σ, $df = n - 1$
Proportion interval	$\widehat{p} \pm z^* \sqrt{\dfrac{\widehat{p}(1-\widehat{p})}{n}}$	Random/independent binary data; success–failure condition
Sample size planning	$n = (z^*\sigma/M)^2$; $n = \widehat{p}(1-\widehat{p})(z^*/M)^2$	Round up; use $\widehat{p} = 0.5$ if no prior estimate
Mean/proportion tests	$t = \dfrac{\bar{x} - \mu_0}{s/\sqrt{n}}$; $z = \dfrac{\widehat{p} - p_0}{\sqrt{p_0(1-p_0)/n}}$	Mean: $df = n - 1$; proportion test uses null-based SE

Responsible communication

Report the data source, sampling or assignment method, procedures used, software or tables used for critical values and p-values, assumptions, uncertainty, and practical context. Reproducible reporting makes statistical claims easier to check.